

The complex numbers

- Introduce a new quantity, i , such that $i^2 = -1$.
- The complex numbers are then all expressions of the form $a + bi$ where a and b are real numbers.

Operations on the complex numbers

- Addition:

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

- Multiplication:

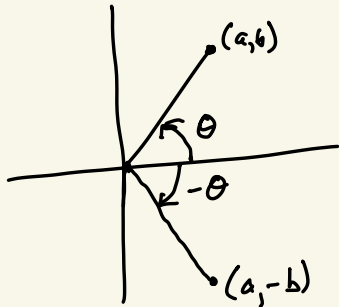
$$(a + bi) \cdot (c + di) = (ac - bd) + (ad + bc)i$$

If $z = a + bi$ then $\bar{z}$, the conjugate of z , is $a - bi$. Note:

$$\underline{z \bar{z} = a^2 + b^2 = |z|^2}$$

If $z \neq 0$ then

$$\frac{1}{z} = \frac{\bar{z}}{z \bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{a - bi}{a^2 + b^2}$$



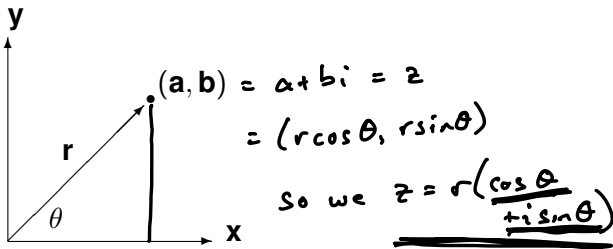
Fundamental Theorem of Algebra (Gauss)

If $p(x)$ is a non-constant polynomial with complex coeff. Then p has a root!

We introduced i as a root of $x^2 + 1$.

We are going to use roots of complex polys as eigenvalues.

The complex plane



- $r = \sqrt{a^2 + b^2}$; this is called the modulus of the complex number $z = a + bi$ and written $|z|$.
- We saw that $z \cdot \bar{z} = |z|^2$.
- θ is an argument for $a + bi$ and is only determined up to multiples of 2π .
- $a = r \cos(\theta)$ and $b = r \sin(\theta)$ so $z = r(\cos(\theta) + i \sin(\theta))$.

A beautiful formula and a caveat going forward

- Exponentiation can be defined on complex numbers as follows:

$$e^{i\theta} = \cos(\theta) + i \sin(\theta)$$

- Consider then that if $\theta = \pi$ then

$$e^{i\pi} + 1 = 0$$

Linear algebra over the complex numbers

Everything we have done in the course to date - linear systems, matrices, determinants, eigenvalues, diagonalization - all goes through **unchanged** if we are using complex numbers instead of real numbers. Going forward it will be considered fair to have complex entries in matrices and to consider complex roots of polynomials.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots$$

Taylor series for e^x

$$e^{i\theta} = 1 + i\theta + \frac{-1 \cdot \theta^2}{2} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \frac{\theta^6}{6!} - \frac{i\theta^7}{7!} + \frac{\theta^8}{8!} + \dots$$

$$\cos \theta \rightarrow \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots \right)$$

$$+ i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots \right)$$

$\uparrow$
 $\sin \theta$

!!
..

Ex: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ Char. poly of A .

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} \\ = \underline{\lambda^2 + 1.}$$

The eigenvalues for A are $\pm i$

Eigenvectors:

$\lambda = i$: Solve $\underline{\begin{pmatrix} -i - 1 & 1 \\ 1 & -i \end{pmatrix}} u = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{aligned} -iu_1 - u_2 &= 0 \\ \underline{u_1 - iu_2} &= 0 \end{aligned}$$

$$u_1 = iu_2 \quad \text{and} \quad u = \underline{t \begin{pmatrix} i \\ 1 \end{pmatrix}}$$

for $t \in \mathbb{C}$

$$\lambda = -i : \text{ Solve } \begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$iu_1 = u_2 \quad \text{and} \quad u = t \begin{pmatrix} 1 \\ i \end{pmatrix} \quad \text{for } t \in \mathbb{C}.$$

$\nearrow$
eigenvectors for $\lambda = -i$.

Multiplication and division in polar form

- If

$$z = r_1 e^{i\theta_1}$$

$$z = r_2 e^{i\theta_2}$$

$$z_1 = \underline{r_1(\cos(\theta_1) + i\sin(\theta_1))} \text{ and } z_2 = \underline{r_2(\cos(\theta_2) + i\sin(\theta_2))}$$

- then $\underline{z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))}$ and
- if $z_2 \neq 0$ then $\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2))$.
- A special case occurs when you compute z^n where $z = r(\cos(\theta) + i\sin(\theta))$:

$$z^n = r^n (\cos(n\theta) + i\sin(n\theta))$$

- We can also compute n^{th} roots of $r(\cos(\theta) + i\sin(\theta))$; they are

$$r^{1/n} \left(\cos\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) + i\sin\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right) \right) \text{ for } k = 0, 1, \dots, n-1$$

Ex:

$$\begin{aligned}\underline{r_1 e^{i\theta_1}} \cdot \underline{r_2 e^{i\theta_2}} &= r_1 r_2 e^{i(\theta_1 + \theta_2)} \\ &= r_1 r_2 \left(\underline{\cos(\theta_1 + \theta_2)} + \underline{i \sin(\theta_1 + \theta_2)} \right)\end{aligned}$$

$$\begin{aligned}& r_1 \left(\underline{\cos \theta_1} + i \underline{\sin \theta_1} \right) \cdot r_2 \left(\underline{\cos \theta_2} + i \underline{\sin \theta_2} \right) \\ &= r_1 r_2 \left(\underline{\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2} \right. \\ & \quad \left. + i \left(\underline{\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2} \right) \right)\end{aligned}$$